DIVYANSHU RAJ

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PROFESSIONAL SUMMARY

Final-year undergraduate in Information Technology (CGPA: 9.29/10) specializing in backend engineering, distributed systems design, and AI-driven applications. Proficient in building high-performance, scalable REST APIs and microservices using Spring Boot, FastAPI, PostgreSQL, and vector databases. Hands-on experience with RAG architectures, LLM integration, and event-driven systems. Strong foundation in DBMS, OS, and cloud deployment using AWS. Seeking backend or software engineering roles to deliver high-availability systems at scale.

TECHNICAL SKILLS

Languages:	Java, Python, SQL
Backend:	Spring Boot, FastAPI, RESTful APIs, Microservices, Spring Security, Spring Data JPA, Hibernate
Frontend:	React.js, Next.js, HTML, CSS, JavaScript
Databases:	PostgreSQL, MySQL, Prisma ORM, Redis
AI / ML:	RAG Systems, LLM Integration, Vector Databases (Pinecone), Whisper, LLaMA
DevOps & Cloud:	Docker, GitHub Actions (CI/CD), AWS, Railway, Kubernetes, Apache Kafka
Tools:	Git, Postman, REST API Testing
Core CS:	Data Structures & Algorithms, DBMS, Operating Systems, OOP, Computer Networks, Agile

PROJECTS

Query Vault 2.0 – **Multimodal AI System** GitHub | Live Demo *FastAPI · PostgreSQL · Pinecone · Next.js*

- Architected a multimodal RAG system processing documents and audio inputs for contextual AI-powered query answering.
- Designed end-to-end ingestion and retrieval pipeline using Pinecone vector database with semantic search and embedding generation; achieved sub-200ms similarity search latency.
- Built high-performance stateless FastAPI backend with concurrent request handling, JWT authentication, and user-level data isolation; optimized for scalable multi-user workloads with low-latency API responses.
- Integrated LLaMA for response generation and Whisper for real-time speech-to-text transcription; modularized ingestion, retrieval, and inference layers for maintainability.
- Deployed full-stack application on Railway with decoupled, modular ingestion pipeline; modular architecture reduced onboarding time for new pipeline components by 40% and supports scalable retrieval and inference workflows.

Task Management System – **Full-Stack Application** GitHub | Live Demo *Spring Boot · React · PostgreSQL*

- Designed RESTful backend using Controller-Service-Repository pattern with optimized PostgreSQL queries and indexing.
- Reduced API response time by 25% through database indexing and query optimization for 10,000+ task records.
- Implemented secure JWT authentication with access and refresh token mechanisms; built CRUD operations, filtering, pagination, and analytics dashboards.
- Improved scalability and maintainability through modular backend design; containerized with Docker for consistent cross-environment deployments; integrated Redis caching layer reducing database round-trips by ~30% and improving API throughput under concurrent load.

Fitness Tracker – **AI-Powered Full-Stack Application** GitHub | Live Demo *Spring Boot · MySQL · React*

- Developed scalable REST APIs using Spring Boot and Spring Data JPA for workout, calorie, and activity management.
- Implemented role-based authentication and authorization using Spring Security and JWT, securing multi-level user data access.
- Integrated AI-powered chatbot delivering personalized fitness and workout suggestions based on user activity and goals.
- Containerized application with Docker for consistent deployments and optimized backend/database performance for efficient activity management.

EDUCATION

KIIT University Bhubaneswar, India
B.Tech in Information Technology May 2026
CGPA: 9.29 / 10

DAV Public School, Cantt Patna Patna, India
Class XII (CBSE): 91.6% | Class X (CBSE): 92.8% 2021 / 2019

CERTIFICATIONS & ACHIEVEMENTS

- **IBM RAG and Agentic AI Professional Certificate** — IBM
- **Runner-up, Lumen Hackathon** — Collaborated in developing a telecom application where I implemented an ML-driven recommendation engine for personalized plan suggestions based on user behavior.